A Note on Rimányi Positivity and the Stronger Laurent Positivity Conjecture for Morin Thom Polynomials

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Abstract

We summarize the positivity conjectures for Thom polynomials of Morin singularities A_d in the form arising from the Bérczi–Szenes residue formula. There are two related statements. The weaker statement, due to Rimányi, predicts nonnegativity of the coefficients of the Thom polynomial when written in the relative Chern classes. The stronger statement, formulated in the Bérczi–Szenes paper, predicts coefficientwise nonnegativity of a Laurent expansion of the rational function appearing in the residue formula. We reproduce a proof of the stronger statement for $d = 4$. For $d = 5$ we find a concrete negative Laurent coefficient, so the stronger statement is false for $d = 5$. This does not disprove Rimányi's Chern-positivity conjecture: several Laurent monomials contribute to the same Chern monomial, and the first negative Laurent coefficient cancels inside a positive Chern coefficient.

1 The residue formula and notation

Let A_d denote the Morin singularity of order d. In the Bérczi–Szenes formula, the Thom polynomial of A_d is expressed as an iterated residue

$$\mathrm{Tp}_{A_d}^{n\to k} = \operatorname*{Res}_{z_1=\infty} \cdots \operatorname*{Res}_{z_d=\infty} \frac{(-1)^d \prod_{1\le m<l\le d}(z_m - z_l)\mathcal{Q}_d(z_1,\dots,z_d)}{\prod_{l=1}^d \prod_{m=1}^{l-1} \prod_{r=1}^{\min(m,l-m)}(z_m + z_r - z_l)} \prod_{l=1}^d c\left(\frac{1}{z_l}\right) z_l^{k-n}\, dz_l. \qquad (1)$$

Here

$$c(t) = 1 + c_1 t + c_2 t^2 + \cdots$$

is the total relative Chern class. We now recall the precise definition of the polynomial $\mathcal{Q}_d$ appearing in the numerator.

The Borel-orbit multidegree defining $\mathcal{Q}_d$

Let

$$T_d \subset B_d \subset GL_d$$

be the diagonal torus and the upper-triangular Borel subgroup. Let $z_1, \dots, z_d$ denote the standard characters of T_d. Consider the GL_d-module

$$\mathrm{Hom}(\mathbb{C}^d, \mathrm{Sym}^2\,\mathbb{C}^d).$$

We write a basis as

$$q_l^{mr} = q_l^{rm}, \qquad 1 \le m, r, l \le d,$$

where q_l^{mr} has T_d-weight

$$z_m + z_r - z_l.$$

Equivalently, q_l^{mr} is the coordinate corresponding to the tensor which sends the lth basis vector of $\mathbb{C}^d$ to the symmetric product of the mth and rth basis vectors.

Define the B_d-invariant linear subspace

$$\widehat{N}_d = \bigoplus_{1 \le m+r \le l \le d} \mathbb{C} q_l^{mr} \subset \mathrm{Hom}(\mathbb{C}^d, \mathrm{Sym}^2 \mathbb{C}^d). \tag{2}$$

Define the reference point

$$\widehat{\epsilon} = \sum_{m=1}^{d} \sum_{r=1}^{d-m} q_{m+r}^{mr} \in \widehat{N}_d. \tag{3}$$

Let

$$\mathcal{O}_d = \overline{B_d \cdot \widehat{\epsilon}} \subset \widehat{N}_d \tag{4}$$

be the Zariski closure of the Borel orbit of this reference point. The polynomial $\mathcal{Q}_d$ is defined to be the T_d-equivariant Poincaré dual, equivalently the T_d-multidegree, of this orbit closure inside the ambient representation:

$$\mathcal{Q}_d(z_1, \dots, z_d) = \mathrm{eP}[\mathcal{O}_d, \widehat{N}_d]_{T_d} = \mathrm{mdeg}_{T_d}(\mathcal{O}_d \subset \widehat{N}_d). \tag{5}$$

Under the standard identification

$$H^*_{T_d}(\widehat{N}_d) \cong H^*_{T_d}(\mathrm{pt}) \cong \mathbb{Z}[z_1, \dots, z_d],$$

this is a homogeneous polynomial of degree

$$\mathrm{codim}_{\widehat{N}_d} \mathcal{O}_d.$$

Computing multidegrees by initial monomial ideals

We also recall the computational definition of the multidegree used in the paper. Let

$$W = \bigoplus_{i=1}^{N} \mathbb{C} y_i$$

be a T-module with T-weight η_i on the coordinate y_i, and let

$$S = \mathbb{C}[y_1, \dots, y_N].$$

If $X \subset W$ is a T-invariant closed subvariety with homogeneous ideal $I_X \subset S$, then

$$\mathrm{mdeg}_T(X \subset W) = \mathrm{mdeg}_T(I_X, S)$$

is the same invariant as the equivariant Poincaré dual $\mathrm{eP}[X, W]_T$. The paper gives an algebraic definition using a finite T-graded free resolution of S/I_X. For explicit calculations one may use the following equivalent algorithmic characterization.

(1) **Flat degeneration to an initial ideal.** Choose a term order or a one-parameter T-compatible Gröbner degeneration and form the initial ideal

$$\mathrm{in}(I_X) \subset S.$$

Since this is a flat degeneration, deformation invariance of the equivariant Poincaré dual gives

$$\mathrm{mdeg}_T(I_X, S) = \mathrm{mdeg}_T(\mathrm{in}(I_X), S). \tag{6}$$

Thus the calculation may be reduced to the case of a monomial ideal.

(2) **Primary decomposition of the monomial ideal.** Let

$$\mathrm{in}(I_X) = \bigcap_a Q_a$$

be a monomial primary decomposition. Keep only the primary components whose radicals

$$P_a = \sqrt{Q_a} = (y_i \mid i \in J_a)$$

have the minimal height, equivalently the components of maximal dimension. Components of smaller dimension do not contribute to the multidegree in the codimension of X.

(3) **Coordinate components.** For a reduced coordinate component

$$L_J = V(y_i \mid i \in J) \subset W,$$

the multidegree is the equivariant Euler class of the normal representation:

$$\mathrm{mdeg}_T(L_J \subset W) = \prod_{i \in J} \eta_i. \tag{7}$$

In words, it is the product of the normal weights.

(4) **Additivity and multiplicities.** By additivity over the maximal-dimensional components, the multidegree of the monomial ideal is

$$\mathrm{mdeg}_T(\mathrm{in}(I_X), S) = \sum_{a:\, \dim V(P_a) = \dim X} m_a \prod_{i \in J_a} \eta_i, \tag{8}$$

where m_a is the scheme-theoretic multiplicity of the primary component Q_a along the coordinate subspace $V(P_a)$. In the square-free reduced case all $m_a = 1$.

Combining (**??**) and (**??**), one obtains a concrete algorithm: compute equations for the T-invariant variety, pass to an initial monomial ideal, find its top-dimensional primary components, and sum the products of the corresponding normal weights, with multiplicities. In the present note this applies to

$$X = \mathcal{O}_d \subset W = \widehat{N}_d,$$

so the weights appearing in (**??**) are the weights

$$\eta(q_l^{mr}) = z_m + z_r - z_l$$

of the normal coordinate directions selected by the relevant coordinate component of the monomial degeneration.

This is the nontrivial orbit-closure input in the Bérczi–Szenes residue formula. All dependence on the Morin order d beyond the universal denominator is encoded in the multidegree $\mathcal{Q}_d$.

The paper points out that, under the convention used for residues at infinity, expanding the denominators in the chamber

$$|z_1| \ll |z_2| \ll \cdots \ll |z_d|$$

cancels the explicit sign $(-1)^d$ in (**??**). Therefore the positivity question may be studied directly from the chamber expansion of the rational function

$$R_d(z_1, \dots, z_d) = \frac{\prod_{1 \le m < l \le d}(z_m - z_l)\mathcal{Q}_d(z_1, \dots, z_d)}{\prod_{l=1}^{d} \prod_{m=1}^{l-1} \prod_{r=1}^{\min(m,l-m)} (z_m + z_r - z_l)}. \tag{9}$$

2 The two positivity conjectures

Conjecture 1 (Rimányi Chern positivity)**.** *For Morin singularities A_d, every coefficient of the Thom polynomial*

$$\mathrm{Tp}_{A_d}^{n \to k}$$

when expanded as a polynomial in the relative Chern classes

$$c_i = c_i(f^*TY - TX)$$

is nonnegative.

Equivalently, if

$$\mathrm{Tp}_{A_d}^{n \to k} = \sum_{\lambda} b_\lambda c_{\lambda_1} \cdots c_{\lambda_s},$$

then Rimányi's conjecture says

$$b_\lambda \ge 0 \quad \text{for every } \lambda.$$

Conjecture 2 (Bérczi–Szenes stronger Laurent positivity)**.** *Expand the rational function R_d from* (**??**) *in the chamber*

$$|z_1| \ll |z_2| \ll \cdots \ll |z_d|.$$

Then the resulting Laurent series has nonnegative coefficients.

Remark 1. *Conjecture* **??** *implies Conjecture* **??***. The reason is that the residue formula extracts Chern monomials from the Laurent expansion of R_d after multiplication by*

$$\prod_{l=1}^{d} c(1/z_l) z_l^{k-n}.$$

If all Laurent coefficients of R_d are already nonnegative, then every Chern coefficient obtained by summing compatible Laurent coefficients is nonnegative. The converse need not hold, because different Laurent monomials can contribute to the same Chern monomial.

3 The case $d = 4$

For $d = 4$, the paper gives

$$\mathcal{Q}_4(z_1, z_2, z_3, z_4) = 2z_1 + z_2 - z_4.$$

Set

$$a = \frac{z_1}{z_2}, \qquad b = \frac{z_2}{z_3}, \qquad c = \frac{z_3}{z_4},$$

so that the chamber $|z_1| \ll |z_2| \ll |z_3| \ll |z_4|$ becomes

$$|a|, |b|, |c| \ll 1.$$

After normalizing by $z_4 = 1$, we have

$$z_3 = c, \qquad z_2 = bc, \qquad z_1 = abc.$$

The rational function becomes

$$F_4(a, b, c) = \frac{(1-a)(1-b)(1-c)(1-ab)(1-bc)(1-abc)(1-bc-2abc)}{(1-2a)(1-2ab)(1-2bc)(1-2abc)(1-b-ab)(1-c-abc)(1-bc-abc)}. \tag{10}$$

Lemma 1. *If f and g are monomials or power series with nonnegative coefficients and zero constant term, then*

$$\frac{1-f}{1-f-g} = 1 + \frac{g}{1-f-g}$$

has nonnegative coefficients in the corresponding small variables.

Proof. The displayed identity is immediate. Since

$$\frac{1}{1-f-g} = \sum_{j \geq 0} (f+g)^j$$

has nonnegative coefficients, so does $1 + g/(1-f-g)$. $\square$

Now factor (**??**) as

$$F_4(a, b, c) = K(a, bc) \cdot \frac{1-b}{1-b-ab} \cdot \frac{1-c}{1-c-abc} \cdot \frac{1-ab}{1-2ab} \cdot \frac{1-abc}{1-2abc}, \tag{11}$$

where

$$K(s, u) = \frac{(1-s)(1-u)(1-u-2su)}{(1-2s)(1-2u)(1-u-su)}. \tag{12}$$

The four factors outside $K(a, bc)$ have nonnegative expansions by Lemma **??**, or by the special case

$$\frac{1-x}{1-2x} = 1 + \frac{x}{1-2x}.$$

It remains to treat K.

A direct identity gives

$$K(s, u) = 1 + \frac{s}{1-2s} + \frac{u}{1-2u} + \frac{s^2 u}{(1-2s)(1-u-su)}. \tag{13}$$

Each term on the right has nonnegative expansion in s, u. Hence $K(s, u)$ is coefficientwise nonnegative.

Combining (**??**) and (**??**), we obtain:

Theorem 1. *The stronger Laurent-positivity conjecture holds for $d = 4$.*

Proof. Every factor in (**??**) has nonnegative expansion in the chamber variables a, b, c. Therefore $F_4(a, b, c)$ has nonnegative coefficients. This is exactly the Laurent expansion of R_4 in the chamber $|z_1| \ll |z_2| \ll |z_3| \ll |z_4|$. $\square$

4 The case $d = 5$: failure of stronger Laurent positivity

For $d = 5$, the paper gives

$$\mathcal{Q}_5(z_1, z_2, z_3, z_4, z_5) = (2z_1 + z_2 - z_5)P_5, \tag{14}$$

where

$$P_5 = 2z_1^2 + 3z_1z_2 - 2z_1z_5 + 2z_2z_3 - z_2z_4 - z_2z_5 - z_3z_4 + z_4z_5.$$

Set

$$a = \frac{z_1}{z_2}, \qquad b = \frac{z_2}{z_3}, \qquad c = \frac{z_3}{z_4}, \qquad e = \frac{z_4}{z_5}.$$

Thus $|z_1| \ll \cdots \ll |z_5|$ becomes

$$|a|, |b|, |c|, |e| \ll 1.$$

Normalize by $z_5 = 1$, so that

$$z_4 = e, \qquad z_3 = ce, \qquad z_2 = bce, \qquad z_1 = abce.$$

The Laurent expansion of R_5 is then an ordinary power series

$$F_5(a, b, c, e) = \sum_{\beta \in \mathbb{Z}^4_{\geq 0}} A_\beta a^{\beta_1} b^{\beta_2} c^{\beta_3} e^{\beta_4}.$$

A finite coefficient computation gives

$$A_{(1,1,2,1)} = [abc^2e]F_5(a, b, c, e) = -1. \tag{15}$$

Since

$$abc^2e = \frac{z_1}{z_2}\frac{z_2}{z_3}\left(\frac{z_3}{z_4}\right)^2\frac{z_4}{z_5} = \frac{z_1z_3}{z_4z_5},$$

this says that the coefficient of the Laurent monomial

$$\frac{z_1z_3}{z_4z_5}$$

in the chamber expansion of R_5 is -1.

Proposition 1. *The stronger Laurent-positivity conjecture is false for $d = 5$.*

Proof. Equation (**??**) exhibits an explicit negative coefficient in the chamber expansion of R_5. Therefore the Laurent expansion is not coefficientwise nonnegative. □

For reference, the coefficient computation behind (**??**) can be organized as follows. Write

$$F_5 = \frac{N_5}{D_5}, \qquad N_5 = \sum_\nu n_\nu a^{\nu_1} b^{\nu_2} c^{\nu_3} e^{\nu_4}, \qquad D_5^{-1} = \sum_\mu h_\mu a^{\mu_1} b^{\mu_2} c^{\mu_3} e^{\mu_4}.$$

Then

$$A_{(1,1,2,1)} = \sum_{\nu \leq (1,1,2,1)} n_\nu h_{(1,1,2,1)-\nu} = -1.$$

This is a finite calculation: only the numerator monomials $\nu \leq (1, 1, 2, 1)$ can contribute.

5 Effect on the weaker Chern-positivity conjecture

The negative Laurent coefficient in Proposition **??** does not imply a negative Chern coefficient. The reason is that the residue formula groups several Laurent monomials into one Chern monomial.

Let

$$\ell = k - n.$$

Suppose

$$a^{\beta_1} b^{\beta_2} c^{\beta_3} e^{\beta_4} = z_1^{\alpha_1} z_2^{\alpha_2} z_3^{\alpha_3} z_4^{\alpha_4} z_5^{\alpha_5}.$$

Then

$$\alpha = (\alpha_1, \dots, \alpha_5) = (\beta_1,\ \beta_2 - \beta_1,\ \beta_3 - \beta_2,\ \beta_4 - \beta_3,\ -\beta_4). \tag{16}$$

The corresponding contribution to the residue is

$$A_\beta \prod_{i=1}^{5} c_{\ell+1+\alpha_i},$$

with the convention $c_j = 0$ for $j < 0$ and $c_0 = 1$.

For the first negative coefficient, $\beta = (1, 1, 2, 1)$, so

$$\alpha = (1, 0, 1, -1, -1).$$

Hence this term contributes negatively to the Chern monomial

$$c_{\ell+2}^2 c_{\ell+1} c_\ell^2.$$

However, it is not the only Laurent monomial contributing to this Chern monomial. One must sum over all orderings of the multiset

$$\{1, 1, 0, -1, -1\}$$

whose partial sums are nonnegative. In this case the nonzero contributions are

α	β	A_β
$(0, 1, 1, -1, -1)$	$(0, 1, 2, 1)$	1
$(1, 0, 1, -1, -1)$	$(1, 1, 2, 1)$	-1
$(1, 1, -1, -1, 0)$	$(1, 2, 1, 0)$	2
$(1, 1, -1, 0, -1)$	$(1, 2, 1, 1)$	2
$(1, 1, 0, -1, -1)$	$(1, 2, 2, 1)$	6

Therefore the actual Chern coefficient is

$$1 - 1 + 2 + 2 + 6 = 10 > 0.$$

Thus the first negative Laurent coefficient cancels inside a positive Chern coefficient.

For example, at relative dimension $\ell = 0$, the residue computation gives the positive Chern expansion

$$\mathrm{Tp}_{A_5}^{\ell=0} = c_1^5 + 10c_1^3 c_2 + 10c_1 c_2^2 + 25c_1^2 c_3 + 12c_2 c_3 + 38c_1 c_4 + 24c_5. \tag{17}$$

All displayed coefficients are positive.

6 Summary of findings

The conclusions are as follows.

(1) The stronger Laurent-positivity conjecture is true for $d = 4$. A direct factorization reduces the claim to elementary positive expansions of the form

$$\frac{1-f}{1-f-g} = 1 + \frac{g}{1-f-g}$$

and to the positive decomposition (**??**).

(2) The stronger Laurent-positivity conjecture is false for $d = 5$. The coefficient of

$$abc^2e = \frac{z_1 z_3}{z_4 z_5}$$

in the chamber expansion is -1.

(3) This negative Laurent coefficient does not disprove Rimányi's original Chern-positivity conjecture. In the first affected Chern monomial, the negative contribution is cancelled by other Laurent monomials, and the resulting Chern coefficient is $10 > 0$.

(4) The present calculation proves the $d = 4$ stronger conjecture and disproves the $d = 5$ stronger conjecture. It also gives an explicit reduction of the $d = 5$ Chern-positivity question to symmetrized sums of Laurent coefficients. A complete proof of Rimányi's weaker Chern-positivity conjecture for all relative dimensions $\ell = k - n$ in the case $d = 5$ would require proving all these symmetrized coefficient sums are nonnegative.